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extensive than it is at present. He *may* be right on this; but I don't think his arguments show that he is.

JOHN WHITE

Scientific Genius: A psychology of science. By Dean Keith Simonton. Pp. 229. Cambridge, Cambridge University Press, 1989. £22.50.

The core of this book is a general theory of scientific creativity. The key idea is that chance permutations of mental elements generate stable configurations that are communicated, socially accepted and preserved socioculturally. This, in effect, is the nature of scientific activity. A scientific genius is a person who does this remarkably well, as judged by public acclaim.

The author does not claim novelty for this theory, but attributes it to Donald Campbell, well known as a metascientist for the application of evolutionary metaphors to scientific discovery. It is also familiar in anecdotal form – for example in a much quoted report by Poincaré of how he came to a certain mathematical idea. The purpose of the book is to develop this theory in detail, demonstrating its applicability to a wide range of observations concerning the psychology of science. The author is not at all coy about this purpose, and insists throughout that this theory is fully confirmed empirically at many points, such as introspective reports, personality and individual differences, the statistics of scientific productivity, the effects of family background and education, and even in the controversial phenomena of multiple discovery.

Indeed, the principal virtue of this work is that it seems very thorough in its review of the diffuse literature on this complex topic. The interpretation of the material evidence – such as it is – is also quite fair, although always directly in terms of the 'chance permutation' theory.

But since the author is so very bullish about his theory, and does not seriously entertain any alternative (which is not all that reprehensible, since there seem to be few coherent enough to dispute the field) we have to ask whether it is plausible in itself. And here one would begin with the whole concept of a 'mental element' in a 'configuration'. Philosophically speaking, this is totally naive. Whatever an 'idea' may be, it is not like a meccano model that can be deconstructed into a finite collection of discrete entities drawn from a denumerable universe. Even a meaningful sentence, even a mathematical formula, has nuances which relate it irrevocably to its origins and context. The intuitive notion that what might possibly appear in such a context may be more or less fluid, more or less diverse, is expressed much too rigidly and narrowly in terms of mere 'permutations'. The positivism of this whole approach takes it from the realm of instructive metaphor into the dreary domain of unconvincing pseudo-quantitative theory.

Even more seriously, there is a very impoverished conception of 'science'. Many thoughtful scientists have argued that the highest talent in science is to perceive, formulate and choose to attack the best problems – 'best' in the sense that Medawar embodied in his characterization of research as 'The Art of the Soluble'. Then there is the research process itself, a highly intentional activity requiring another sort of expertise, perhaps a different type of personality. After the supposed moment of 'discovery', there is usually a long period of testing where many would-be scientists come to grief through inadequate self-criticism. Even the writing up and dissemination of the results has its own problems and perils.

What Professor Simonton fails to appreciate is that every stage of this process has

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its characteristic difficulties. The supposed 'elements' that would need to come together to make progress at one stage would be quite different in kind and provenance from those required at another. And yet they would be linked together: the original appreciation of a scientific 'problem' would be a source for, *and would decisively modify*, the 'mental elements' going into the research process and beyond. It is quite unrealistic to suppose that even scientific geniuses have a fixed set of such elements acquired at some much earlier period in their scientific lives on which they draw at random to deal with every new eventuality.

Indeed, if we consider the rapid progress of scientific knowledge in every field, we can see that the constitutents of the discoveries that a scientist makes later in life must include numerous items that were simply not present at all at an earlier stage in his or her career. One of the features of consistently high scientific achievement is just this ability to *learn* new things, new ways of thought, from year to year, and from decade to decade. In any case, these and all the other constituents of our thoughts are formed and reshaped by the society in which we live. Professor Simonton's reluctance to accept even a weak cultural factor in his model is somewhat perverse.

As an academic psychologist he also surprisingly ignores recent studies which indicate the important role of 'mental models' in cognitive processes. It would not be difficult to develop a theory involving deliberate cogitation of various aspects of such a model, turning it around in imagination, so to speak, searching for new ways of understanding it. Some scientific achievements, surely, depend upon a superior capacity for the synthesis of such models and their orderly analysis.

Perhaps because he has devoted his own scientific career to the study of 'creativity', Professor Simonton wants to assimilate all manifestations of this phenomenon to the same theoretical explanation. In one sense, it is certainly true that scientific research is an 'art', not susceptible to complete rational explication. But that is not to say that artistic genius, in its usual sense, is just the same sort of thing as scientific genius. As the author points out in his peculiar attempt to ward off the threat posed by multiple discoveries, it is too simple to suppose that there is a predetermined 'solution' for the scientist to 'discover'. But the task of the scientist is made just that bit more challenging by the fact that it is inexorably bounded and framed by established truths and unavoidable realities that are far stronger than the conventions and traditions that the artist may respect, but may in the end boldly overstep.

The main interest of this book is its very fascinating subject. It contains bits and pieces of information that qualify for storage and random retrieval in quite other contexts. There is also a certain amount of truth in its central idea that a fertile mind with a low threshold for wild ideas can be a major asset in the hard graft of research. But the earnest endeavour to ennoble this cliché into a formula is unconvincing. There are occasions when scientific obtuseness is even more inexplicable than scientific genius!

JOHN ZIMAN

The Education Crisis: Which Way Now? By Norman Graves. Pp. xi, 154. London: Christopher Helm, 1988. £9.95.

It is difficult to know which readers are addressed in this book, for essentially it is a brisk statement of one man's view of the English educational system at present, surveying its various aspects – structure, nursery and primary education, secondary, further and continuing education, higher education, tests and examinations – in what is avowedly a personal way, referring in passing to his own experiences, since the

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